

Program 8

Create Kubernetes Pods using YAML Descriptors

DevOps

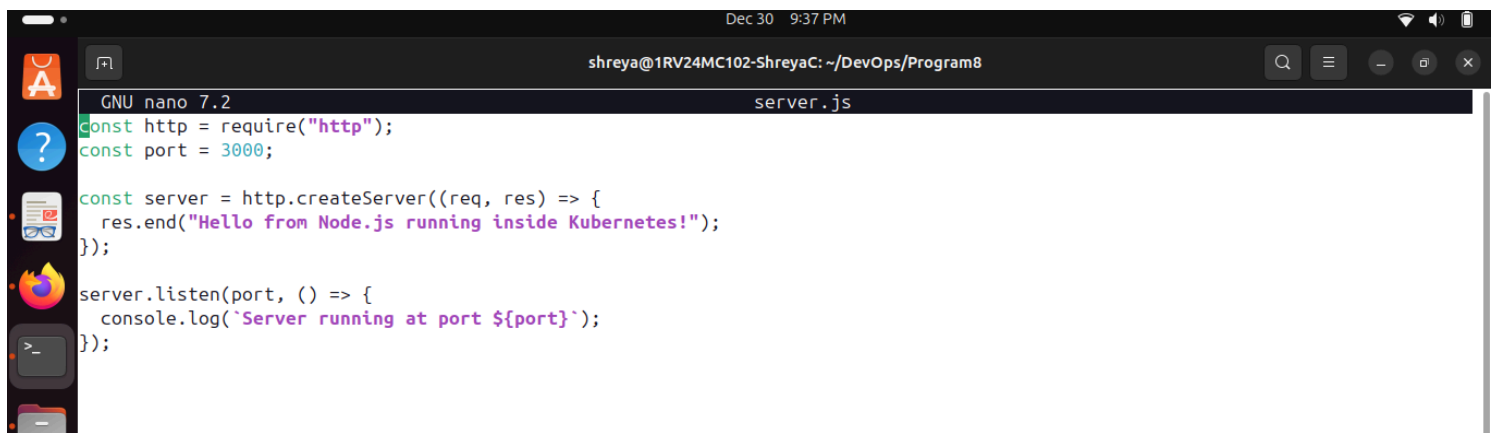
— Program8

— nodejs-configmap.yml

— nodejs-pod.yml

— [server.js](#)

Step 1 : Create [server.js](#) file and add the following code

A screenshot of a terminal window titled 'shreya@1RV24MC102-ShreyaC: ~/DevOps/Program8'. The terminal shows the GNU nano 7.2 editor editing a file named 'server.js'. The code in the file is as follows:

```
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 2 : Create nodejs-configmap.yml and add the following code

A screenshot of a terminal window titled 'shreya@1RV24MC102-ShreyaC: ~/DevOps/Program8'. The terminal shows the GNU nano 7.2 editor editing a file named 'nodejs-configmap.yml'. The code in the file is as follows:

```
=apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

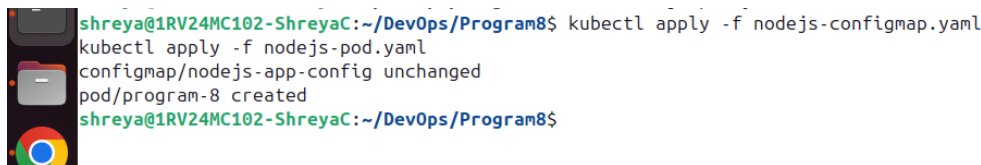
    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: Create nodejs-pod.yml and add the following code



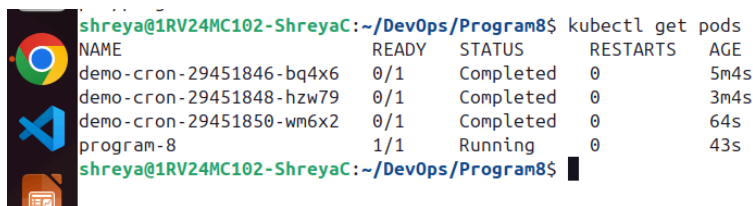
```
GNU nano 7.2 nodejs-pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
  - name: nodejs
    image: node:18
    command: ["node", "/usr/src/app/server.js"]
    volumeMounts:
    - name: app-code
      mountPath: /usr/src/app
    ports:
    - containerPort: 3000
  volumes:
  - name: app-code
    configMap:
      name: nodejs-app-config
```

Step 4 : Apply the YAML files



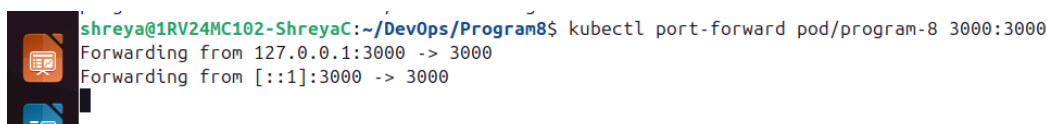
```
shreya@1RV24MC102-ShreyaC: ~/DevOps/Program8$ kubectl apply -f nodejs-configmap.yaml
kubectl apply -f nodejs-pod.yaml
configmap/nodejs-app-config unchanged
pod/program-8 created
shreya@1RV24MC102-ShreyaC:~/DevOps/Program8$
```

Step 5 : Check the pod status



```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
demo-cron-29451846-bq4x6            0/1     Completed 0           5m4s
demo-cron-29451848-hzw79            0/1     Completed 0           3m4s
demo-cron-29451850-wm6x2            0/1     Completed 0           64s
program-8                           1/1     Running   0           43s
shreya@1RV24MC102-ShreyaC:~/DevOps/Program8$
```

Step 6 : Access the [node.js](#) app



```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

